Unit-1:Introduction To Gen Ai

Course Outcomes & Syllabus

- **CO1: Explain fundamentals, evolution and types of generative AI models.**

Syllabus: Introduction To Gen Ai: Historical Overview of Generative modeling, Difference between Gen AI and Discriminative Modeling, Importance of generative models in AI and Machine Learning, Types of Generative models, GANs, VAEs, autoregressive models and Vector quantized Diffusion models, understanding of probabilistic modeling and generative process, Challenges of Generative Modeling, Future of Gen AI, Ethical Aspects of AI, Responsible AI, Use Cases.

T1: Denis Rothman, "Transformers for Natural Language Processing and Computer Vision", Third Edition,Packt Books, 2024

R1: David Foster," Generative Deep Learning", O'Reily Books, 2024.

R2: Altaf Rehmani, "Generative AI for Everyone", BlueRose One, 2024

Learning Outcomes

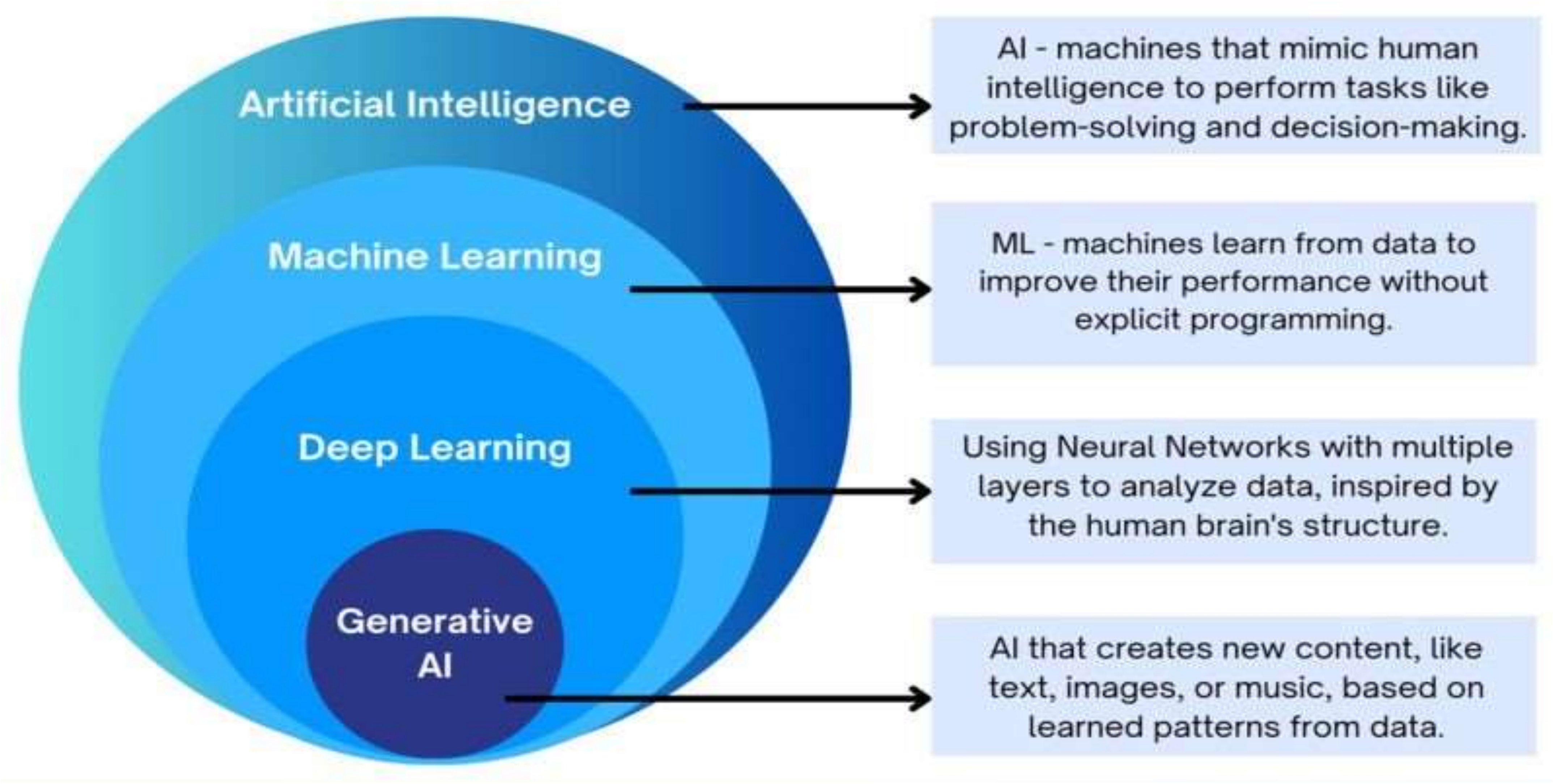
At the end of this lecture, Student will be able to:

LO1:Understand the fundamentals, evolution and importance of Generative models in AI and Machine Learning.

LO2:Analyze different Generative AI models such as GANs, VAEs, Autoregressive Models, and Diffusion Models with their applications

LO3: Evaluate the challenges, ethical aspects, responsible AI practices, and future trends of Generative AI systems.

AI,ML,DL and Gen AI



AI,ML,DL and Gen AI

What is AI (Artificial Intelligence)?

Artificial Intelligence (AI) is a technology that enables machines to think and act like humans.

What is Machine Learning (ML)?

Machine Learning is a part of AI where computers learn from data and improve their performance automatically without being explicitly programmed for each task.

Types of Machine Learning:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

AI,ML,DL and Gen AI

Types of Machine Learning:

1.Supervised Learning:-

Supervised Learning is a method where a computer learns from labeled data and uses this knowledge to make accurate predictions on new data. Example: Predicting house prices from size, location and number of Rooms

2.Unsupervised Learning:-

Unsupervised Learning is a method where a computer examines unlabeled data and identifies hidden patterns, relationships, or groups on its own. Example: Grouping customers by shopping habits.

3.Reinforcement Learning:-

Reinforcement Learning is a method where a computer learns by interacting with an environment, receiving rewards or penalties, and gradually discovering the best actions to take. Example: A robot learning to walk, or AI playing games.

*Deep Learning

It is a special type of Machine Learning that uses **neural networks** with many layers. It is very powerful for tasks like: Image recognition (e.g., recognizing faces)

Introduction To Gen AI

- **Generative AI** refers to **artificial intelligence models that can create new content**, such as text, images, music, code, or even video, by learning patterns from existing data.

- These models don't just analyze data-they **generate** something **original** that resembles what they've been trained on.

Examples :

- Write an essay or story (e.g., Chat GPT)
- Create a digital painting (e.g., DALL·E, Mid journey)
- Compose a song or melody
- Generate realistic human speech (e.g., voice cloning)
- Auto-generate code or documentation

Historical Overview of Generative Modeling

1950s–1980s: Foundations in Statistics:

Early generative models such as Gaussian Mixture Models (GMMs) and Naive Bayes were developed. These models aimed to learn the joint probability distribution of input data and output labels.

1950s–1980s: Foundations in Statistics

Early probabilistic generative models learned the joint probability distribution of data.

Examples:

Naive Bayes (1950s): Probabilistic text and email classification.

Gaussian Mixture Models (1970s): Data clustering and synthetic data generation.

Hidden Markov Models (1970s): Speech and handwriting generation/recognition.

Bayesian Networks (1980s): Probabilistic reasoning and expert systems.

Historical Overview of Generative Modeling

1990s–2000s: Probabilistic Graphical Models:

Development of models like Hidden Markov Models (HMMs), Bayesian Networks, and Latent Dirichlet Allocation (LDA).These provided more structured ways to model uncertainty and dependencies in data.

Period	Generative Model	Year	Example Application
1990s	**Hidden Markov Model (HMM)**	Widely used in the 1990s	Speech recognition, handwriting recognition, bioinformatics
1990s	**Bayesian Networks**	Popularized in the 1990s	Medical diagnosis, expert systems, fault detection
2003	**Latent Dirichlet Allocation (LDA)**	2003	Topic modeling, document clustering, recommendation systems

Historical Overview of Generative Modeling

2014: Breakthrough with Generative Adversarial Networks (GANs):

Ian Good fellow introduced Generative Adversarial Networks (GANs).

GANs use a generator and a discriminator in a competitive setup to create highly realistic synthetic data, especially images. This innovation marked a major turning point in the field of generative AI.

GAN Model	Year	Example Use
Deep Convolutional GAN (DCGAN)	2015	Realistic image generation
CycleGAN	2017	Convert horses to zebras, summer to winter images
Pix2Pix	2017	Sketch-to-photo, map-to-satellite image generation
StyleGAN	2018	Photorealistic human face generation
StyleGAN2	2019	High-quality synthetic face generation

Historical Overview of Generative Modeling

2017–2020: Deep Generative Models Expanded:

Introduction and growth of models such as Variational Auto encoders (VAEs), Pixel CNN, and Transformer-based models. These models expanded generative capabilities across **text, images, audio, and beyond**.

Period	Generative AI Model	Introduced	Example Application
2017–2020	**Variational Autoencoder (VAE)**	2013–2014 (widely adopted later)	Image generation, anomaly detection, data compression
2017–2020	**Pixel CNN**	2016	High-quality image generation and image completion
2017–2020	**Transformer**	2017	Language translation, text generation, summarization
2018–2020	**GPT**	2018	Text generation, question answering, dialogue systems
2018–2020	**BERT**	2018	Text understanding, search, question answering

Historical Overview of Generative Modeling

2020s–Present: Generative AI Goes Mainstream:

The rise of large-scale models like GPT-3, GPT-4, DALL·E, Stable Diffusion, and Chat GPT. Generative AI becomes widely used in fields such as education, healthcare, design, entertainment, and programming.

1. **GPT-3 (2020)**
   - Large Language Model capable of generating coherent, human-like text.
   - **Example:** Essay writing, chat bot development, programming assistance.
2. **Chat GPT (2022)**
   - Conversational AI built on GPT models.
   - **Example:** Virtual tutoring, customer support, coding help, content creation.
3. **DALL·E 2 (2022)**
   - Generates high-quality images from natural language descriptions.
   - **Example:** Logo design, illustrations, marketing graphics.
4. **Stable Diffusion (2022)**
   - Open-source diffusion model for creating and editing realistic images.
   - **Example:** Digital artwork, product visualization, image in painting.
5. **GPT-4 (2023)**
   - Advanced multimodal Large Language Model capable of understanding text and images.
   - **Example:** Research assistance, software development, document analysis, educational tutoring.

Difference between Gen AI and Discriminative Modeling

Aspect	Generative AI (Generative Modelling)	Discriminative Modelling
Definition	Models that learn the **joint probability** P(x, y) or P(x)P(x)	Models that learn the **conditional probability** P(y)
Purpose	Generate new data similar to the training data	Classify or predict labels from input data
What it learns	Both how data is distributed and how labels relate to features	Direct relationship between input features and output labels
Data Generation	Can **generate** realistic new samples (e.g., images, text, music)	Cannot generate new data
Examples	GANs, VAEs, Diffusion Models, Naive Bayes, GPT	Logistic Regression, SVM, Decision Trees, Random Forest, Neural Networks

Difference between Gen AI and Discriminative Modeling

Output	Synthetic data (e.g., images, text, audio)	Class labels, predictions
Use Cases	Text generation, image synthesis, creative AI, data augmentation	Classification, regression, and object detection
Training Complexity	Usually more complex and computationally intensive	Comparatively simpler and faster to train
Interpretability	Often less interpretable	More interpretable in many traditional models
AI Focus	Focuses on **creation** and **imagination**	Focuses on **decision-making** and **label prediction**

Importance of Generative Models in AI & ML

1. Learning Data Distribution

❖Generative models learn how data is distributed, capturing the structure and patterns in datasets.

❖This understanding is useful for building smarter systems that can mimic real data.

2. Data Generation

❖They can generate new, realistic data samples, useful for training, testing, or creative applications.

❖For example, generating images, text, or even tabular data that resembles real-world inputs.

3. Handling Limited or Unlabeled Data

❖Useful in semi-supervised and unsupervised learning, where labeled data is scarce or limited.

❖They can learn from and generate new examples to improve training effectiveness.

4. Creative Applications (AI Focus)

❖Enables natural language generation, image synthesis, music composition, etc.

❖Powers tools like Chat GPT, DALL·E, and other generative AI applications.

Importance of Generative Models in AI & ML

5. Anomaly Detection (The process of identifying rare data points, events, or observations that deviate significantly from standard or expected patterns)

By Modelling normal data patterns, generative models can identify outliers or anomalies effectively.

6. Simulation and Decision Support

- ❖Used in AI for simulating future scenarios, predicting possible outcomes, or imagining environments.
- ❖Helps in tasks like planning and reinforcement learning.

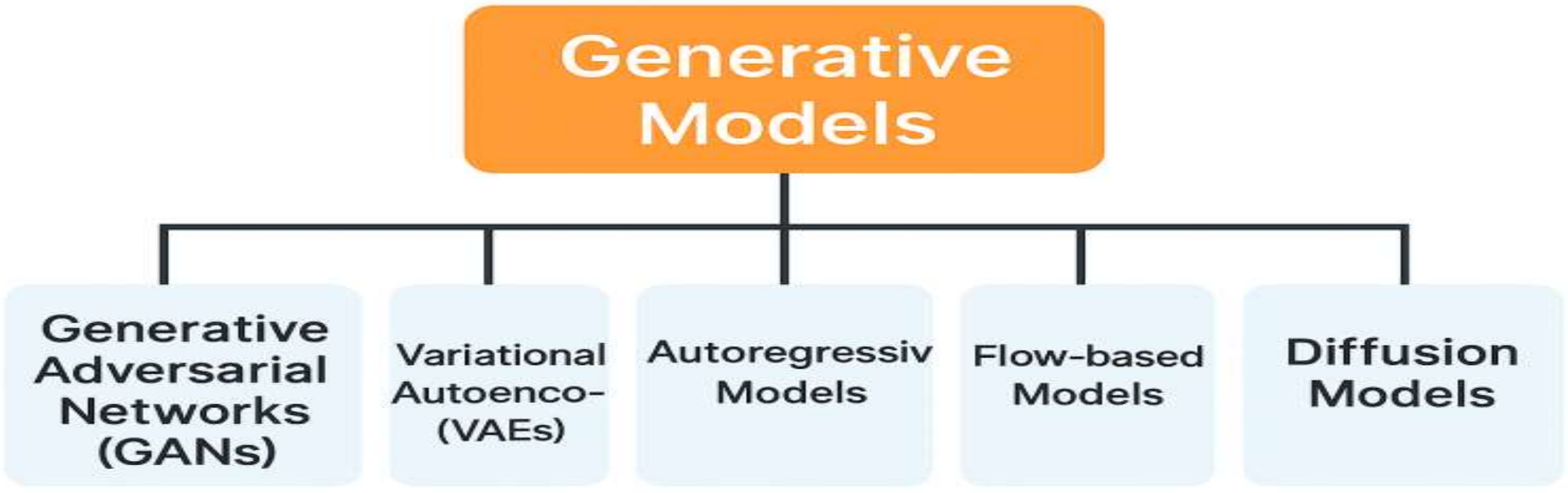
7. Multimodal Learning

- ❖Supports learning across multiple data types—like combining text and images (e.g., caption generation).
- ❖Useful in systems that need to interpret and generate across different media.

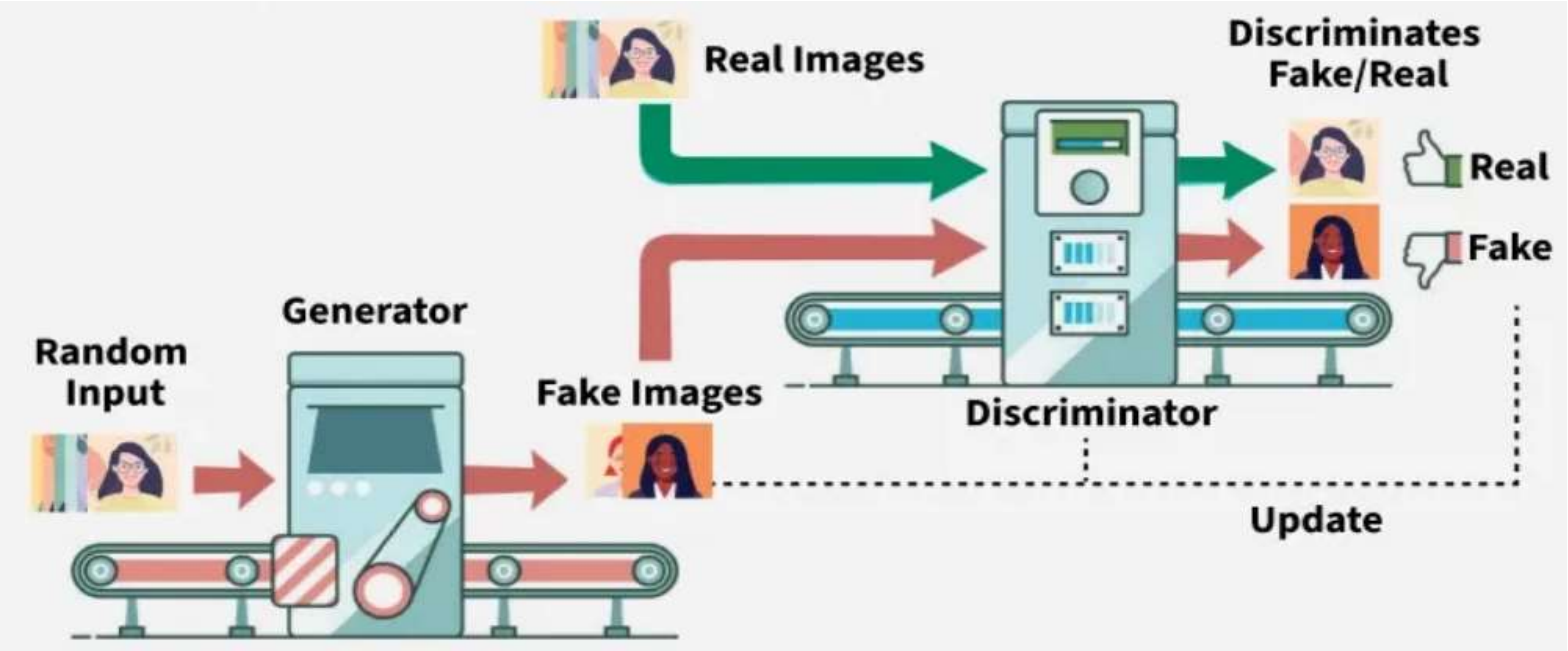
8. Foundation for Advanced Models

- ❖Underpins many advanced AI architectures: GANs, VAEs, diffusion models, etc.
- ❖Essential for developing flexible, adaptive, and generative AI systems.

Types of Generative Models



GAN (Generative Adversarial Network)



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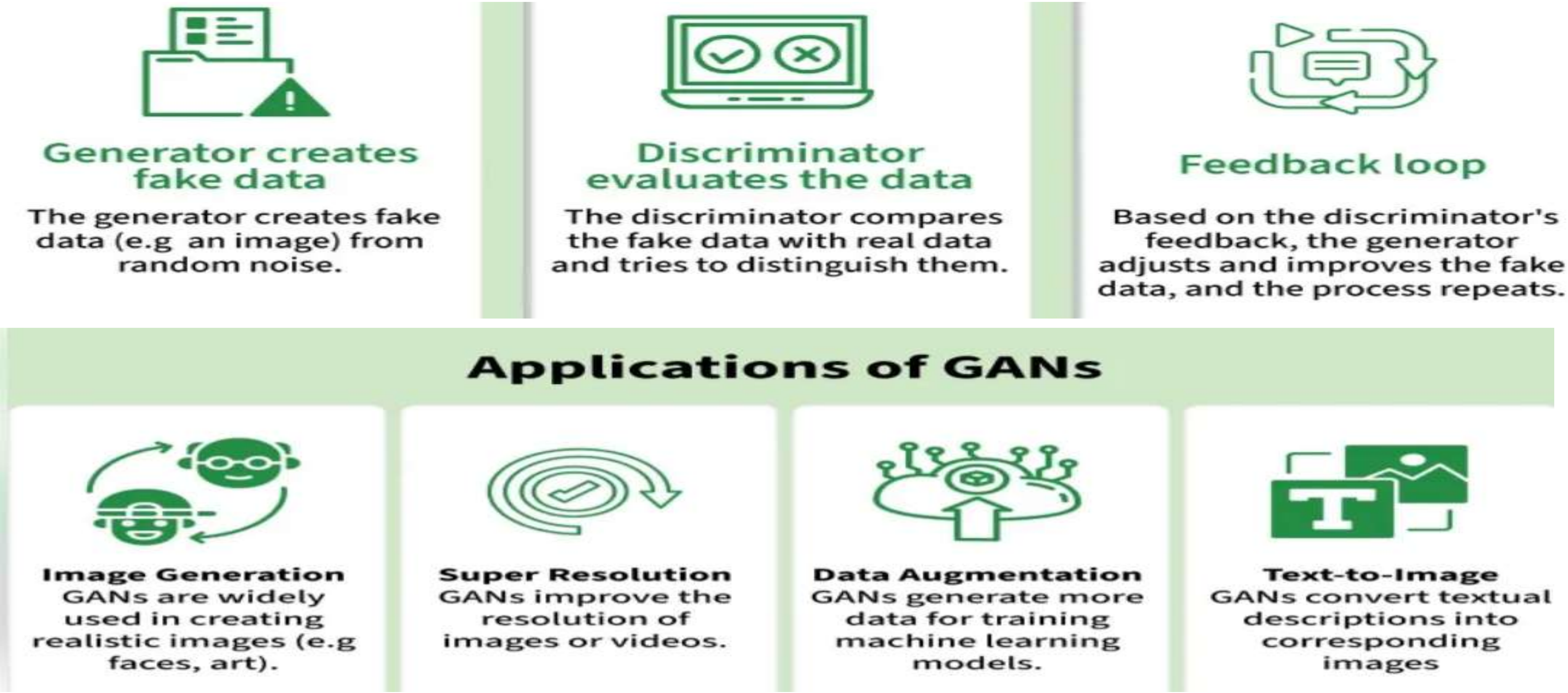

GAN (Generative Adversarial Network)

GAN (Generative Adversarial Network)

- Introduced in **2014 by Ian Good fellow.**
- GANs **do not require labelled data** to train.
- The model learns to generate data by **learning the distribution** of the input data.
- It only needs **samples from the real data** (e.g., images) and **random noise** to train the Generator.
- GANs train two networks (Generator and Discriminator) using **real data without labels**. That's why they are an **unsupervised** branch of **Machine Learning.**

GANs consist of **two competing neural networks:**

Generator (G):

Takes random noise as input.

Tries to **generate fake data** that looks like real data.

Learns to **fool the Discriminator**.

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GAN (Generative Adversarial Network)

Discriminator (D):

Takes real or fake data as input.

Learns to **classify data as real or fake**.

Helps the Generator improve by giving feedback

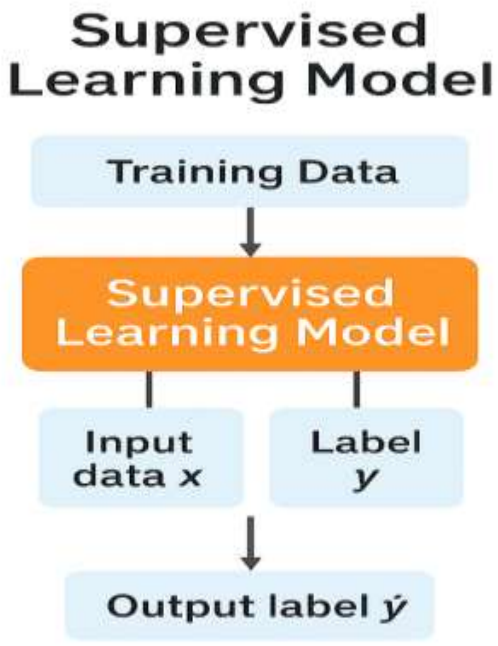
Working Steps of a GAN

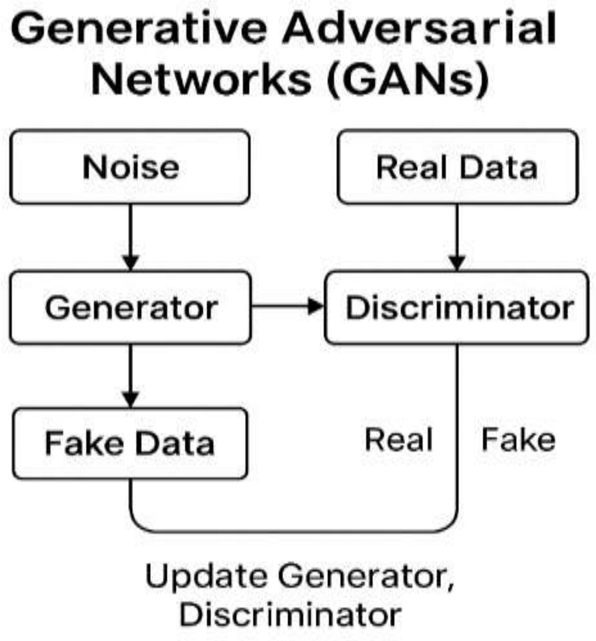
1. **Input Random Noise (z):** This noise is fed into the **Generator**.
2. **Generator Creates Fake Data (G(z)):** The Generator uses the noise to create **fake data samples** (e.g., fake images).
3. **Discriminator Evaluates Real vs. Fake:**
   - The Discriminator receives both:
     - **Real data samples** from the training dataset.
     - **Fake data samples** from the Generator.
   - It tries to **classify them** as real (1) or fake (0).

GAN (Generative Adversarial Network)

Use Cases for GANs

- **Image generation** (e.g., faces, landscapes).
- **Data augmentation** (especially for training small datasets).
- **Synthetic medical data** to train AI without privacy risks.
- **Fashion design**, **art creation**, and **deep fake detection**.





Variational Auto encoder (VAE)

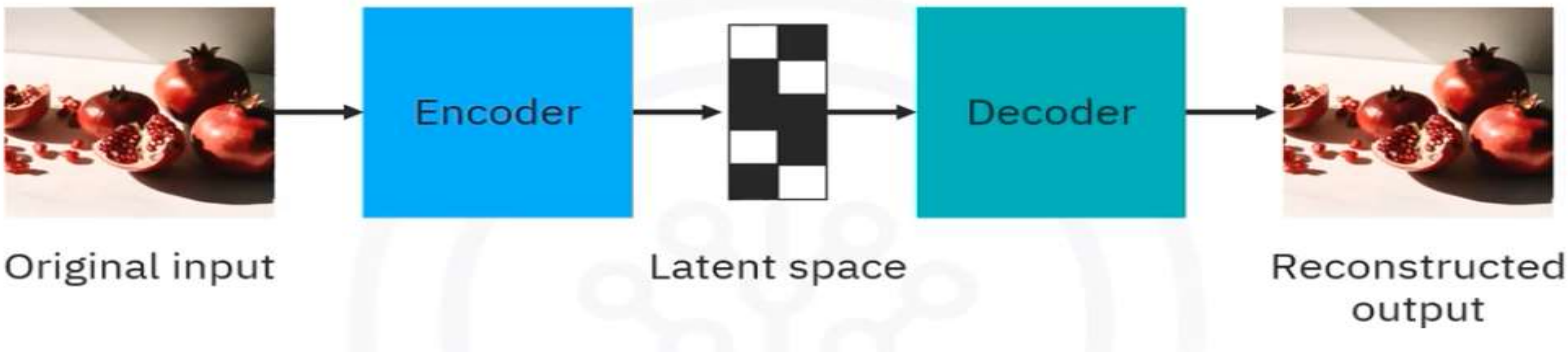
A VAE is a type of neural network that learns to compress data into a structured latent space and can **generate new, realistic data** by sampling from that space.

Key Features

1.Versatile Data Handling: training data, making them suitable for a wide range of use.

2.Dimensionality Reduction: They reduce the dimensionality of data, which helps in creating improved versions of the original data.

3.Probabilistic Model: VAEs utilize probabilistic methods to learn the underlying distribution of the data.



Mechanism

VAEs consist of two main components: an encoder and a decoder.

1.Encoder:

- The encoder is a neural network that processes the input data to identify its most important features.
- It compresses the data into a lower-dimensional representation, known as the latent space.
- This involves learning the probability distribution of the data and isolating key variables.
- The compressed data is stored in the latent space, which serves as a compact representation of the input data.

Variational Auto encoder (VAE)

2.Latent Space:

- The latent space is a mathematical space within the model's architecture where the compressed data is stored.
- It represents the data in a reduced-dimensional format, making it easier to manipulate and generate new samples.

3.Decoder:

- The decoder is another neural network that takes the compressed representation from the latent space and reconstructs it into the original data format.
- It performs the reverse operation of the encoder, aiming to generate data that closely resembles the original input.
- The training process involves minimizing the difference between the input data and the reconstructed output using a maximum likelihood principle.

Variational Auto encoder (VAE)

Training Process

- **Maximum Likelihood Principle:** VAEs are trained to minimize the difference between the original input and the reconstructed output, ensuring the generated data is as close to the original as possible.
- **Continuous Latent Space:** Despite being trained in a static environment, the latent space in VAEs is continuous, allowing for the generation of new data samples by randomly sampling from the learned distribution.

Applications

1. Image Synthesis
2. Data Compression
3. Anomaly Detection
4. Industry Examples:
    - Entertainment
    - Finance
    - Health care

Autoregressive Models

An **Autoregressive (AR) Model** is a generative model that **predicts the next element** in a sequence **based on previous elements.** It builds sequences **step by step**, where each prediction depends on the outputs generated so far.

Structure of an AR

Step-by-step flow:

1. **Start with a prompt** (e.g., first word or image pixel).
2. **The model predicts the next value** using previous values. (RNN, Transformer, etc.,)
3. **Add the prediction to the sequence.**
4. Repeat the process to build the full output.

Autoregressive Equation:

$$P(x) = P(x_1) \cdot P(x_2|x_1) \cdot P(x_3|x_1, x_2) \cdot \ldots \cdot P(x_n|x_1, \ldots, x_{n-1})$$

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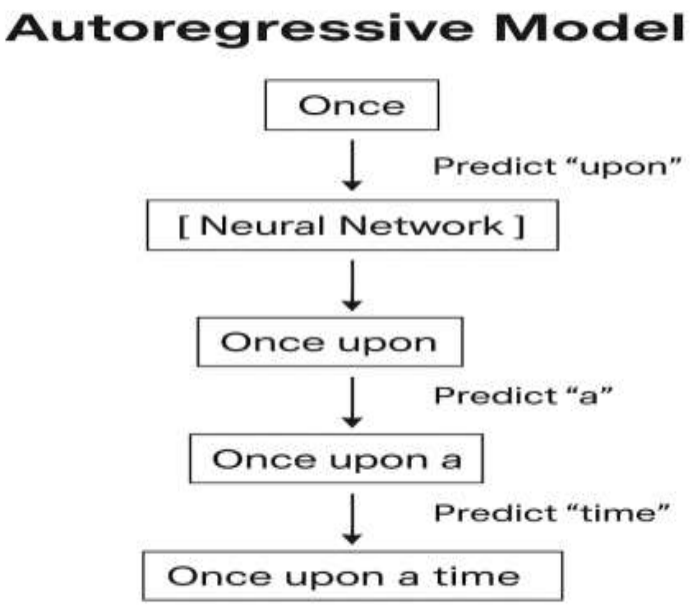
Autoregressive Models



Example: Word-by-Word Storytelling

Prompt: You say "Today,"

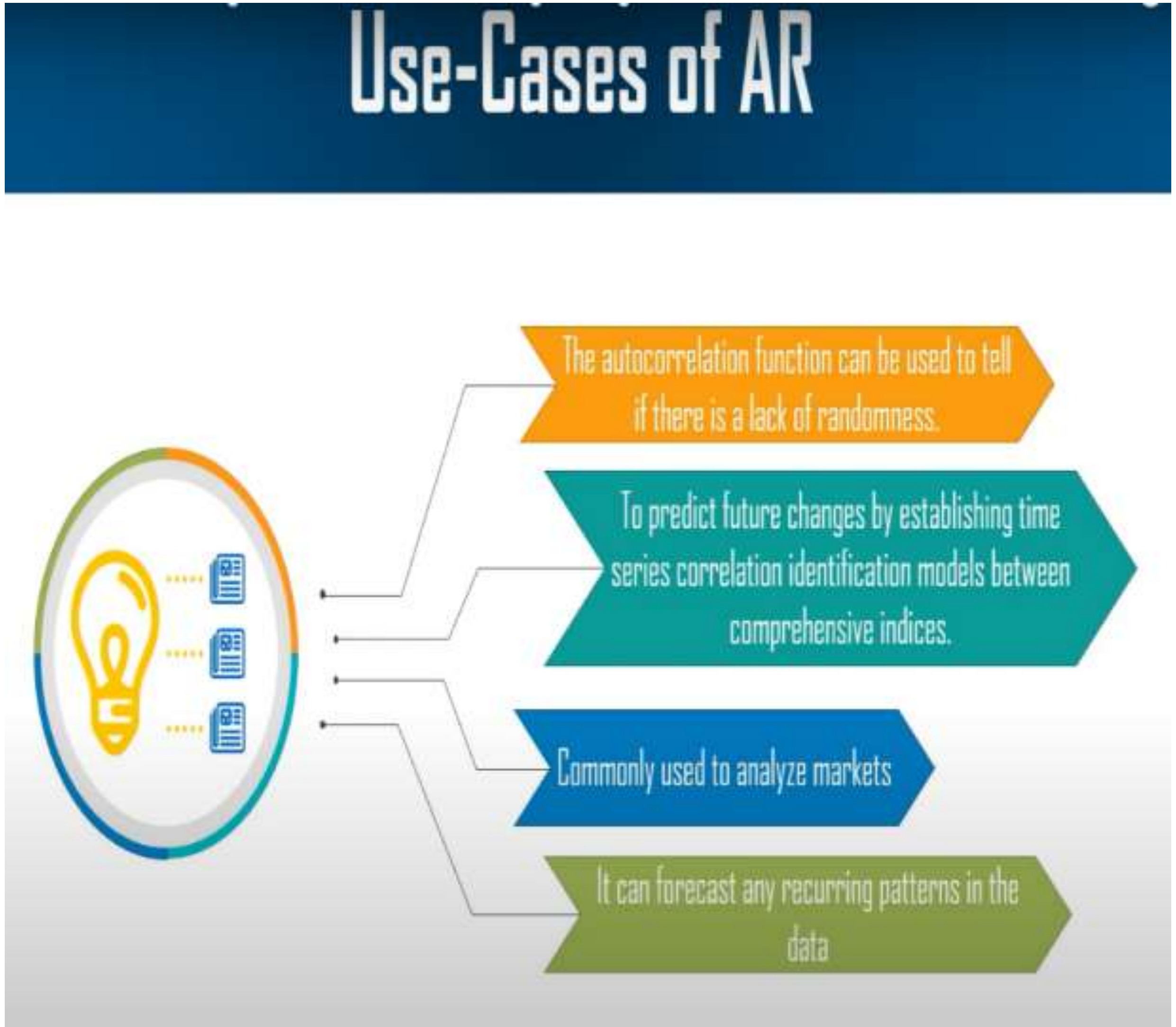
- **Model predicts**: "I"

Because "Today I" is a common phrase based on its training.

- The sentence so far: "Today I"
- **Model predicts**: "went"
  - Now your sentence is "Today I went"
- **Model predicts next**: "to"
- Then: "the"
- Then: "park."

Resulting in: **"Today I went to the park."**

Autoregressive Models



Domain	Example Models	What They Generate
Text	GPT, LLaMA	Sentences, code, answers
Audio	Wave Net	Speech, music
Images	Pixel RNN, Pixel CNN	Pixel-wise image generation
Music	Music Transformer	Notes, compositions

Vector quantized Diffusion models

What is a Diffusion Model?

A **Diffusion Model** is a deep learning model that learns to generate new images by **removing noise step by step**.

Simple Analogy

Imagine you have a clear photograph.
First, you gradually add noise until it becomes complete random static.
Then you train a neural network to reverse this process.
Once the model learns how to remove noise, it can start from random noise and gradually create a realistic image.1.**Forward Diffusion Process,2.Reverse Diffusion Process** and 3.**Conditional Diffusion**

Vector Quantization (VQ)

Vector Quantization (VQ) is a technique that converts continuous vectors into discrete vectors (tokens) by replacing each vector with its nearest representative vector from a predefined codebook.

Vector quantized Diffusion models

Vector Quantized Diffusion Model

A Vector Quantized Diffusion Model (VQ-Diffusion) is an image generation model that combines two powerful techniques:

Vector Quantization (VQ) – Compresses an image into discrete tokens.

Diffusion Model – Learns to generate these discrete tokens by gradually removing noise.

1. Forward Diffusion Process

The **forward process** gradually adds noise to the data. After many steps, the data becomes pure Gaussian noise.

2. Reverse Diffusion Process

The **reverse process** is learned by a neural network to gradually remove noise from a random noisy sample, and generate new data (like an image, audio, etc.).

Vector quantized Diffusion models

3. Conditional Diffusion

In **conditional diffusion**, we guide the reverse process using **extra information** like text prompts, class labels, or images.
For example, in **text-to-image generation**, the condition is a **text prompt** like "a cat wearing sunglasses".

Text Prompt: "A cat wearing sunglasses"

↓

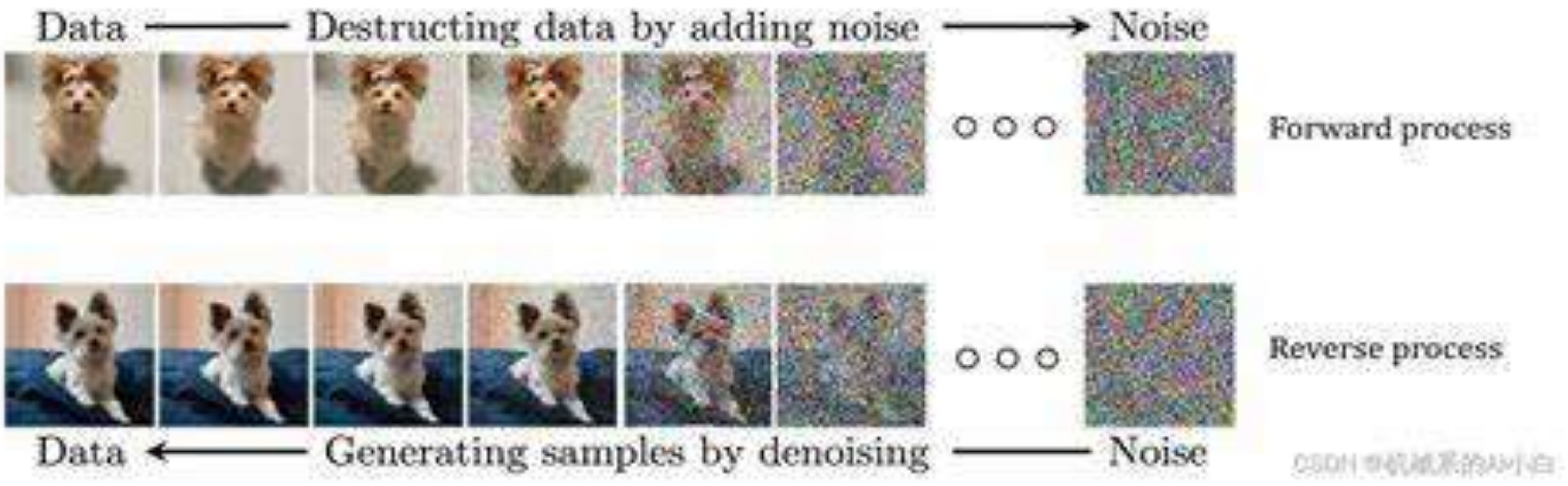
Conditioning input

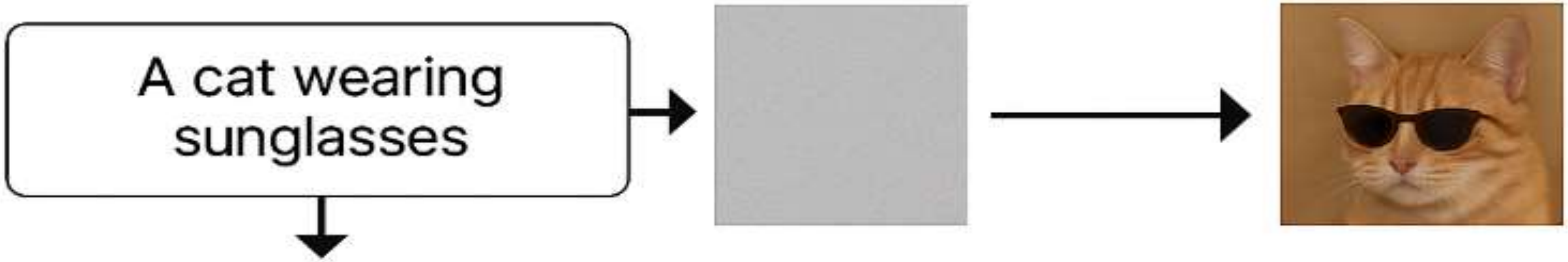
↓

Noise → Model(x_t | text) → Denoised Step

→ ... repeat ... → Final Image 🐱🕶

Vector quantized Diffusion models





Understanding Probabilistic Modelling and the Generative Process

Probabilistic Modeling is a technique that predicts how likely different outcomes are by using probability..

It tells us how likely different outcomes are, given our observations.

Imagine you're predicting tomorrow's weather.

- You don't say: "It **will** rain."
- You say: "There's a **70% chance** of rain."

That's **probabilistic reasoning.**

A generative process is a method of creating new data by learning patterns from existing data.

Example:

Learn from thousands of cat images.

Understand common features such as ears, eyes, and tail.

Generate a **new cat image** that has never existed before.

Understanding Probabilistic Modeling and the Generative Process

Why is This Useful?

- Uncertainty-aware predictions
- Generates realistic data (text, images, speech)
- Core of **Generative AI**, including **ChatGPT**, **DALL·E**, **Stable Diffusion**

```
DATA  →   Learn P(x)  →  Use P(x) to Generate New Data
              ↑                   ↓
        Probabilistic        Generative
          Modeling              Process
```

Understanding Probabilistic Modelling and the Generative Process

Model	Type	Description
VAE	Probabilistic	Learns latent space + reconstructs data
Diffusion	Probabilistic	Adds/removes noise using distributions
Autoregressive (e.g., GPT)	Probabilistic	Predicts the next token using conditional probabilities
GANs	Not fully probabilistic	Adversarial training, less explicit probability

Challenges of Generative Modeling

- **Mode Collapse:** The model repeatedly generates similar outputs instead of diverse samples.
- **Training Instability:** Training may become unstable, causing slow or failed convergence.
- **High Computational Cost:** Training and generating data require significant computing power and time.
- **Likelihood Estimation:** Accurately estimating the probability of complex data is difficult.
- **Evaluation is Tricky:** There is no single metric to reliably measure the quality of generated content.
- **Controlling Output:** Precisely controlling the generated content remains challenging.

Challenges of Generative Modeling

- **Bias and Ethical Issues:** Biased training data can produce unfair, misleading, or harmful outputs.
- **Safety and Misuse:** Generative AI can be misused to create deepfakes, fake content, and misinformation.
- **Multimodal Learning Challenges:** Combining text, images, audio, and video effectively is still a complex task.
- **Over fitting/Memorization:** The model may memorize training data instead of learning general patterns.

Future of Gen AI

- **Multimodal AI:** Future models will seamlessly understand and generate text, images, audio, video, and 3D content together.
- **Personalized AI:** AI systems will generate content tailored to individual users' preferences and needs.
- **Scientific Discovery:** Generative AI will accelerate research in drug discovery, materials science, and healthcare.
- **Creative Collaboration:** AI will assist humans in creating art, music, code, games, and digital content more efficiently.
- **Responsible and Ethical AI:** Future AI will focus on fairness, transparency, privacy, and reducing bias and misinformation.
- **Real-Time AI Applications:** Faster models will enable real-time content generation for education, healthcare, robotics, and autonomous systems.

Future of Gen AI

- **Smaller and Efficient Models:** Future generative models will require less memory and computational power while maintaining high performance.
- **AI Agents:** Autonomous AI agents will perform complex tasks by reasoning, planning, and interacting with tools and environments.
- **Domain-Specific AI:** Specialized generative models will be developed for medicine, engineering, finance, law, and education.
- **Human-AI Collaboration:** Generative AI will become an intelligent assistant that enhances human creativity and productivity rather than replacing humans.

Ethical Aspects of AI

Ethics in AI deals with ensuring that AI technologies are **fair**, **safe**, **transparent**, and **respectful of human rights**.

Key Ethical Issues:

Ethical Concern	Description
Bias & Fairness	AI may reinforce existing biases from data (e.g., gender or racial bias).
Privacy	Use of personal data raises privacy risks (e.g., facial recognition).
Accountability	Who is responsible when AI systems cause harm or error?
Transparency	Black-box models lack explainability. Users need to understand decisions.
Autonomy	AI should not undermine human decision-making or freedom.
Safety & Security	Malfunctioning or adversarial attacks on AI can cause harm.
Job Displacement	AI can automate jobs, affecting employment and social stability.

Responsible AI

Responsible AI ensures that AI is developed and used in a way that aligns with ethical principles and societal values.

Core Principles of Responsible AI:

Principle	Description
Fairness	Avoid discrimination and ensure equitable outcomes.
Transparency	Clearly explain how AI models work and make decisions.
Accountability	Assign responsibility to developers and organizations.
Privacy Protection	Safeguard user data and give control to individuals.
Robustness	Ensure reliability, accuracy, and resistance to manipulation.
Human-Centered	Design AI to enhance, not replace, human roles.
Inclusiveness	Engage stakeholders from diverse backgrounds in AI design.

Use Cases of Responsible AI

Sector	Use Case	Description
Healthcare	AI Diagnostics	AI helps detect diseases (e.g., cancer, diabetic retinopathy) ethically.
Finance	Fraud Detection	AI identifies suspicious transactions while protecting user privacy.
Education	Adaptive Learning Systems	Personalized learning while ensuring data protection and inclusion.
Agriculture	Smart Farming	AI predicts crop yield, optimizing resources without harming communities.
Government	Public Policy Analysis	AI supports data-driven decisions transparently and fairly.
Environment	Climate Modeling	AI forecasts environmental changes and promotes sustainability.
Law & Justice	Predictive Policing	Used cautiously to avoid racial profiling or unfair targetin

Use Cases of Responsible AI

Best Practices for Implementing Ethical AI:

- Conduct **AI impact assessments**.
- Include **interdisciplinary teams** (ethics, law, tech).
- Use **auditable** and **interpretable models**.
- Apply **data minimization** and **differential privacy** techniques.
- Train models on **diverse, representative datasets**.

Suggested Tools & Frameworks:

- AI Fairness 360 (IBM)
- Microsoft Responsible AI Standard
- Google's AI Principles
- EU's AI Act and Ethics Guidelines
- IEEE Ethically Aligned Design

Summary

•Explained the fundamentals, historical evolution, and significance of Generative AI, including generative and discriminative modeling concepts.

•Covered major Generative AI models such as GANs, VAEs, Autoregressive Models, and Vector Quantized Diffusion Models with their working principles and applications.

•Discussed challenges, future trends, ethical aspects, responsible AI practices, and real-world use cases of Generative AI.

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Thank You